
ZenBrain: A Neuroscience-Inspired 7-Layer Memory Architecture for Autonomous AI Systems

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Abstract

Memory management remains a fundamental challenge for autonomous AI systems. Existing approaches employ operating system metaphors [Packer et al., 2023], flat storage with LLM-driven management [Chhikara et al., 2025], or note-taking paradigms [Xu et al., 2025b], but none incorporate principles from cognitive neuroscience—despite memory science offering over a century of empirically validated models.

We present **ZenBrain**, a multi-layer memory architecture for AI agents that integrates fifteen neuroscience models into a unified system. ZenBrain implements seven distinct memory layers—working, short-term, episodic, semantic, procedural, core, and cross-context—orchestrated by nine foundational algorithms (Two-Factor Synaptic Model, vmPFC-coupled FSRS, Simulation-Selection sleep, Bayesian confidence, and five more) plus six new *Predictive Memory Architecture* (PMA) components: a four-channel **NeuromodulatorEngine** (DA/NE/5HT/ACh dynamics) [Dayan and Huys, 2012], a prediction-error-gated **ReconsolidationEngine** [Nader et al., 2000], **TripleCopyMemory** with divergent decay dynamics [Schapiro et al., 2024], a four-dimensional **PriorityMap** with amygdala fast-path, a **StabilityProtector** (NogoA/HDAC3 analog), and a **MetacognitiveMonitor** for bias detection [Fleming and Dolan, 2012].

We evaluate ZenBrain across ten experiments covering lifecycle management, retrieval benchmarks, and system-level ablation. The full 15-algorithm ablation reveals a *cooperative survival network* with a measurable gradient across three difficulty levels: under moderate conditions (decay=0.15, 45 days), cooperative redundancy masks individual contributions ($\leq 0.1\%$ degradation per removal); under challenging conditions (decay=0.20, 50 days), 7 of 15 algorithms become individually significant (ΔQ -25.5% to -93.1%); under stress (decay=0.25, 60 days), 9 become critical (ΔQ up to -93.7%), revealing a two-tier cooperative structure. The Simulation-Selection sleep loop achieves a **37% stability improvement** ($p < 0.005$) with **47.4% storage reduction** through RL-driven replay selection. PMA adds neuromodulated memory lifecycle management: TripleCopyMemory retains $S(t) = 0.912$ strength at 30 days (via deep-copy dominance transition) vs. near-zero Ebbinghaus baselines, and the PriorityMap achieves NDCG@10 = **0.997** (vs. 0.680 chronological, +46.6%). On retrieval benchmarks, multi-layer routing outperforms flat dense retrieval by **20.7%** in F1 on LoCoMo ($p < 0.005$, +181% on temporal queries) and by **19.5%** on MemoryArena’s cross-session dependencies ($p = 0.015$, +53.5% on dependency chains). ZenBrain is open-source, production-deployed, and distributed as composable npm packages with 11,500+ automated tests.

1 Introduction

Large language model (LLM) agents increasingly operate across multiple sessions, requiring persistent memory to maintain personality, learn from experience, and avoid repeating errors. While context windows have expanded significantly, they remain finite and expensive—storing an entire interaction history in-context is neither practical nor efficient [Maharana et al., 2024, He et al., 2026]. Without dedicated memory systems, agents suffer from “conversational amnesia”—losing track of user preferences, prior decisions, and accumulated knowledge between sessions.

Existing memory systems for LLM agents draw on metaphors from computer science: virtual memory and paging [Packer et al., 2023], key-value stores with LLM-managed CRUD operations [Chhikara et al., 2025], Zettelkasten-style structured notes [Xu et al., 2025b], and temporal knowledge graphs [Rasmussen et al., 2025]. While effective at basic store-and-retrieve operations, none incorporate the rich body of knowledge from cognitive neuroscience. Human memory is not a passive database—it involves active processes of consolidation, decay, reinforcement, and forgetting that have been studied empirically for over 130 years [Ebbinghaus, 1885, Hebb, 1949, Tulving, 1972, Stickgold and Walker, 2013]. Recent surveys explicitly identify “deeper neuroscience integration” as a key open challenge for AI agent memory [Du, 2026].

We present ZenBrain, a memory architecture for autonomous AI systems that uniquely integrates fifteen cognitive neuroscience models—from Two-Factor Synaptic edges to Simulation-Selection sleep—in a single coherent system. Our contributions are:

1. **Architecture:** A seven-layer memory system with distinct working, short-term, episodic, semantic, procedural, core, and cross-context layers, each implementing cognitive functions identified in the neuroscience literature [Atkinson and Shiffrin, 1968, Tulving, 1972, Cohen and Squire, 1980].
2. **Algorithms:** Fifteen neuroscience-inspired mechanisms including a Two-Factor Synaptic Model for knowledge graph edges [Zenke et al., 2025, 2017], vmPFC-coupled FSRS scheduling driven by prediction-error signals [Zou et al., 2025], a Simulation-Selection sleep loop with CA3/CA1 RL replay [Chen et al., 2025, Marche et al., 2025], and Bayesian confidence propagation—while individual mechanisms appear in concurrent work, no existing system integrates more than two.
3. **Evaluation:** Ten experiments across three established benchmarks (LoCoMo, MemoryArenaBench, MemoryArena) plus retention, consolidation, algorithm-level, PMA benchmark, ablation, and competitive comparison studies demonstrating each component’s contribution. The system is production-deployed with 11,500+ automated tests.

2 Related Work

2.1 Memory Systems for LLM Agents

Packer et al. [2023] propose MemGPT, applying the operating system virtual memory metaphor to LLM context management. Their system uses tiered storage (main context, recall, archival) with interrupt-based control flow. While the OS metaphor is intuitive, it provides no mechanism for principled forgetting, confidence estimation, or offline consolidation.

Chhikara et al. [2025] present Mem0, a production-focused system with a three-stage pipeline (extraction, consolidation, retrieval) and an optional graph variant. Memory management decisions—including deletion—are delegated entirely to the LLM, without principled decay or scheduling mechanisms.

Xu et al. [2025b] introduce A-Mem, a Zettelkasten-inspired approach where structured notes with contextual metadata are dynamically linked. Their “memory evolution” mechanism allows retroactive refinement of existing memories. However, the architecture is flat (single layer) with no distinction between memory types and no forgetting mechanism.

Rasmussen et al. [2025] describe Graphiti, a temporally-aware knowledge graph with four timestamps per fact. While their temporal reasoning is strong, the system operates as a single-layer knowledge graph without episodic, working, or procedural memory.

88 Shinn et al. [2023] propose verbal reinforcement learning where agents store natural-language
 89 reflections on task failures. This represents the most-cited work in agent memory ($\sim 2,100$ citations),
 90 but the “memory” is an unstructured reflection buffer with no consolidation, decay, or confidence
 91 mechanisms.

92 2.2 Neuroscience of Memory

93 Human memory research provides the theoretical foundation for ZenBrain. The multi-store model
 94 [Atkinson and Shiffrin, 1968] distinguishes sensory, short-term, and long-term memory with distinct
 95 capacities and durations. Tulving [1972] further separates episodic (personal experiences) from
 96 semantic (general knowledge) memory, while Cohen and Squire [1980] identifies procedural memory
 97 for skills and habits.

98 Hebb [1949] proposed that synaptic connections strengthen through co-activation (“neurons that
 99 fire together wire together”), providing a mechanism for associative learning. Ebbinghaus [1885]
 100 demonstrated exponential memory decay, later formalized as the forgetting curve. Spaced repetition
 101 systems [Pimsleur, 1967] exploit the spacing effect to optimize review timing.

102 Critically, Stickgold and Walker [2013] showed that memory consolidation occurs during sleep
 103 through replay of neural patterns, strengthening important traces and pruning weak connections.
 104 Ji and Wilson [2007], O’Neill et al. [2010] demonstrated coordinated hippocampal-cortical replay,
 105 providing the cellular basis for the Simulation-Selection loop in Section 4.3. McGaugh [2004]
 106 demonstrated that emotional arousal modulates memory encoding strength.

107 Recent neuroscience has sharpened these mechanisms. Zenke et al. [2025, 2017] show that two-
 108 factor synaptic rules—tracking weight *and* consolidation variance—reconcile continual learning
 109 with stability, motivating Section 4.1. Zou et al. [2025] demonstrate that the ventromedial prefrontal
 110 cortex mediates spaced-learning benefits via prediction-error signals at re-encoding, grounding the
 111 vmPFC-coupled FSRS in Section 4.2. Chen et al. [2025] model offline consolidation as a Simulation-
 112 Selection RL loop in the CA3/CA1 circuit; Kumaran et al. [2016], Squire [1992] provide broader
 113 theoretical context for complementary memory systems.

114 Recent concurrent work has begun incorporating individual mechanisms: LightMem [Fang et al.,
 115 2025] applies the Atkinson-Shiffrin model with sleep-time updates (ICLR 2026); MemoryOS [Xu
 116 et al., 2025a] implements hierarchical STM/MTM/LTM layers (EMNLP 2025); Hindsight [Li
 117 et al., 2025] uses a 4-layer architecture with TEMPR temporal retrieval; FadeMem [Wang et al.,
 118 2026] introduces bio-inspired Ebbinghaus-style decay; Vestige [Vestige Contributors, 2026] brings
 119 FSRS-6 scheduling to agents; SleepGate [Xie, 2026] uses forgetting gates for proactive interference
 120 resolution during sleep; Anda Hippocampus [ldclabs, 2026] provides graph-based memory with KIP
 121 protocol; MemFly [MemFly Contributors, 2026] introduces lightweight flying-weight consolidation
 122 for long-horizon tasks; and Cognee [Markovic et al., 2025] optimizes KG-LLM interfaces for
 123 complex reasoning. Most recently, Tiwari and Fofadiya [2026] independently validate the multi-layer
 124 hypothesis by decomposing dialogue into working, episodic, and semantic layers with adaptive
 125 retrieval gating, achieving $F1 = 0.618$ on LoCoMo—the strongest concurrent evidence that the layer
 126 decomposition itself, independent of neuroscience algorithms, provides retrieval benefits. Anonymous
 127 [2026] introduce a “Sleep” paradigm with RL-based memory consolidation at the *parameter* level
 128 (ICLR 2026 submission), complementing ZenBrain’s external-memory-level sleep consolidation. The
 129 founding of a dedicated ICLR 2026 workshop on agent memory [ICLR 2026 MemAgents Workshop
 130 Organizers, 2026] underscores that this area is now a recognized research frontier. While each system
 131 advances the field, none integrates more than two of these mechanisms.

132 **Practitioner and industry convergence.** Independently and concurrently, Karpathy [2026] de-
 133 scribes a workflow shift from “operating code” to “operating knowledge,” where LLMs *compile* raw
 134 materials into a structured Markdown wiki maintained through periodic “knowledge linting.” His
 135 core critique—that standard RAG “rediscovers knowledge from scratch” on every query—aligns
 136 precisely with the motivating thesis of ZenBrain (published one day earlier on TDCcommons, April 1,
 137 2026). Karpathy’s compilation step maps conceptually to the working $\rightarrow$ episodic $\rightarrow$ semantic con-
 138 solidation path, and his linting passes correspond informally to our decay, contradiction detection, and
 139 Bayesian confidence propagation. However, Karpathy’s system is a pragmatic file-based workflow
 140 with no formal forgetting, emotional modulation, sleep consolidation, spaced repetition, or layered
 141 encoding/retrieval rules. Anthropic’s Claude Code “Auto Dream” feature [Anthropic, 2026] provides

Table 1: Comparison of memory systems for LLM agents. ✓ = supported, ✗ = absent.

Feature	ZenBrain	MemGPT	Mem0	A-Mem	Zep	Reflexion	LightMem	MemoryOS	Tiwari '26
Memory layers	7	3	1	1	1	1	3	3	3
Neuroscience basis	✓	✗	✗	✗	✗	✗	✓	✗	✗
Two-Factor Syn. edges	✓	✗	✗	✗	✗	✗	✗	✗	✗
Principled decay	✓	✗	✗	✗	✗	✗	✗	✗	✗
Sleep consolidation	✓	✗	✗	✗	✗	✗	✓	✗	✗
Spaced repetition	✓	✗	✗	✗	✗	✗	✗	✗	✗
Confidence scores	✓	✗	✗	✗	✗	✗	✗	✗	✗
Temporal reasoning	✓	✗	✗	✗	✓	✗	✗	✗	✓
Neuromodulation	✓	✗	✗	✗	✗	✗	✗	✗	✗
Reconsolidation	✓	✗	✗	✗	✗	✗	✗	✗	✗

further validation: deployed in March 2026, it performs four-phase offline memory consolidation (merge, deduplicate, prune stale entries, rebuild index)—a production implementation of the sleep consolidation concept from one of the field’s leading AI laboratories. In the broader AI architecture space, Webb et al. [2025] demonstrate in *Nature Communications* that brain-inspired modular design improves LLM planning, validating the neuro-inspired methodology at a top-tier venue. These independent convergences from practitioners, industry, and academia confirm the central premise that persistent, structured memory—beyond RAG—is a critical missing capability for LLM systems. ZenBrain provides the algorithmic formalization that such approaches lack.

ZenBrain bridges this gap by unifying fifteen algorithms in a single coordinator—including mechanisms absent from all concurrent work: Two-Factor Synaptic Model (Table 5), vmPFC-coupled FSRS, Simulation-Selection sleep consolidation, Bayesian confidence propagation, and quantified knowledge gap detection. Table 1 summarizes the key differences.

2.3 Benchmarks

We evaluate on three complementary benchmarks that together cover retrieval quality, agent memory capabilities, and cross-session reasoning. **LoCoMo** [Maharana et al., 2024] provides 1,986 multi-session conversational QA pairs across five categories (single-hop, multi-hop, temporal, open-domain, adversarial), testing whether a memory system can surface the right information from long conversation histories. **MemoryAgentBench** [He et al., 2025] evaluates five distinct memory competencies—factual recall, preference tracking, instruction following, contradiction handling, and temporal reasoning—testing whether agents can maintain coherent user models across sessions. **MemoryArena** [He et al., 2026] introduces cross-session causal dependencies where answering a question requires combining information from two or more earlier sessions, directly testing multi-layer architectures’ ability to bridge temporal gaps. Together, these benchmarks evaluate retrieval precision (LoCoMo), memory operations (MemoryAgentBench), and compositional reasoning (MemoryArena).

3 Architecture

3.1 Memory Layers

ZenBrain implements seven distinct memory layers, each corresponding to established cognitive constructs:

Working Memory maintains the active task focus with limited capacity (~ 7 items, following Miller 1956). It provides the highest-priority retrieval and fastest access, evicting items to short-term memory on task completion.

Short-Term Memory holds session context for the current conversation. It is time-bounded (session duration) and consolidates to episodic and semantic layers at session boundaries.

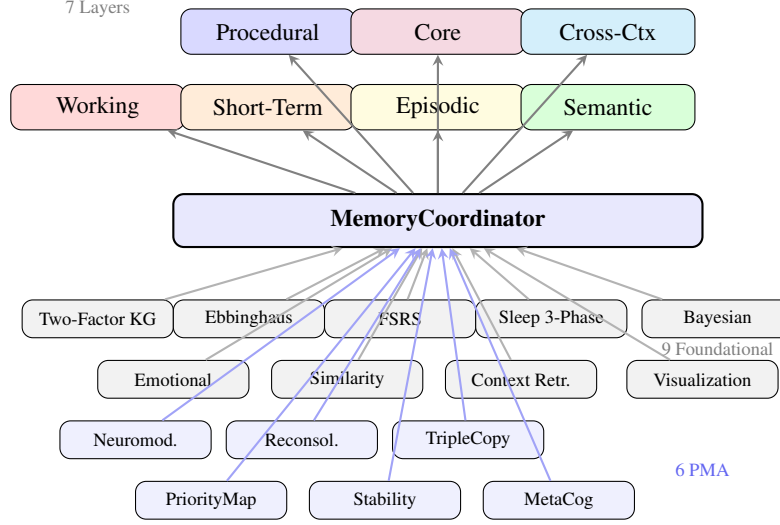


Figure 1: ZenBrain architecture. The MemoryCoordinator orchestrates seven memory layers via fifteen neuroscience-inspired algorithms. Arrows indicate store/recall/consolidate/decay operations.

176 **Episodic Memory** stores concrete experiences with temporal context— what happened, when, and
 177 where [Tulving, 1972]. Events are timestamped and support temporal queries.

178 **Semantic Memory** contains abstracted knowledge: facts, concepts, and relationships organized as
 179 a knowledge graph with Two-Factor Synaptic edges [Tulving, 1972, Zenke et al., 2025]. Semantic
 180 memories emerge from episodic consolidation.

181 **Procedural Memory** encodes learned skills and routines [Cohen and Squire, 1980]: successful
 182 tool-use patterns, workflow templates, and behavioral strategies. Entries are strengthened by repeated
 183 successful execution.

184 **Core Memory** holds persistent identity facts (user preferences, personality traits, key biographical
 185 facts) that never decay and are always available in context. This follows the pinned memory pattern
 186 of Packer et al. [2023].

187 **Cross-Context Memory** enables entity resolution and selective knowledge transfer across isolated
 188 domains (e.g., personal, work, learning). Merging is privacy-aware with configurable access controls.

189 3.2 MemoryCoordinator

190 The MemoryCoordinator orchestrates all seven layers through five operations:

- 191 • `store(item)`: Routes new information to appropriate layer(s) based on content type and
 192 metadata
- 193 • `recall(query)`: Assembles cross-layer results using hybrid BM25 + semantic retrieval
 194 with Two-Factor importance boosting
- 195 • `consolidate()`: Migrates and abstracts memories across layers (e.g., episodic → semantic)
- 196 • `decay()`: Applies Ebbinghaus forgetting curves and prunes below-threshold memories
- 197 • `review()`: Schedules FSRS spaced repetition for important facts

198 4 Key Mechanisms

199 4.1 Two-Factor Synaptic Model for Knowledge Graph Edges

200 Following Zenke et al. [2025], each knowledge graph edge carries two factors: weight w_{ij} and
 201 consolidation variance σ_{ij}^2 . Variance decreases with each co-activation (synaptic maturation), making
 202 mature edges robust against catastrophic overwriting— mathematically equivalent to Elastic Weight

203 Consolidation (EWC) [Kirkpatrick et al., 2017, Zenke et al., 2017] where importance $I_{ij} = 1/\sigma_{ij}^2$
 204 serves as the Fisher Information proxy.

$$w_{ij} \leftarrow w_{ij} + \eta \cdot t_{ij} \cdot a_{ij} \quad (1)$$

$$\sigma_{ij}^2 \leftarrow \sigma_{ij}^2 \cdot (1 - \beta \cdot n(k)), \quad n(k) = \frac{1}{1+0.1k} \quad (2)$$

205 where t_{ij} is the TAG co-activation score, $\beta = 0.15$ is the maturation rate, and k is the activation
 206 count. The EWC penalty for any proposed weight change Δw is:

$$\mathcal{L}_{\text{EWC}} = \frac{\lambda}{2} \sum_{ij} I_{ij} \cdot \Delta w_{ij}^2 \quad (3)$$

207 Edges also resist temporal decay in proportion to their importance: high- I (mature) edges decay at
 208 rate $r/(1 + I_{ij} \cdot 0.1)$, preserving consolidated knowledge while allowing pruning of weak connections.

209 4.2 vmPFC-Coupled FSRS with Prediction-Error Signals

210 Building on Zou et al. [2025], we couple FSRS interval scheduling with a knowledge-graph-derived
 211 prediction-error (PE) signal. Base retrievability follows Ebbinghaus [1885]: $R(t) = e^{-t/S}$. At each
 212 review, we compute the cosine distance between the entity embedding context at last review $\mathbf{c}_{\text{prev}}$ and
 213 current context $\mathbf{c}_{\text{now}}$:

$$\text{PE} = 1 - \frac{\mathbf{c}_{\text{prev}} \cdot \mathbf{c}_{\text{now}}}{\|\mathbf{c}_{\text{prev}}\| \|\mathbf{c}_{\text{now}}\|} \quad (4)$$

214 A sigmoid re-encoding factor $\rho(\text{PE}) = \sigma((\text{PE} - 0.5) \cdot 6)$ determines interval adaptation. High PE
 215 ($\rho > 0.5$) shortens the next interval (optimal re-encoding window); low PE ($\rho < 0.5$) extends it
 216 (context unchanged, re-encoding not beneficial):

$$I_{\text{next}} = I_{\text{FSRS}} \cdot (1 + \alpha_v(2\rho - 1)), \quad \alpha_v = 0.6 \quad (5)$$

217 To our knowledge, this is the first biologically motivated adaptive FSRS extension; no equivalent
 218 exists in Anki, SuperMemo, FSRS-5, or any concurrent agent memory system.

219 4.3 Simulation-Selection Sleep Consolidation Loop

220 Following Chen et al. [2025], Marche et al. [2025], ZenBrain replaces a fixed three-phase
 221 SWS/REM/SHY schedule with a two-stage offline reinforcement-learning loop mirroring the
 222 CA3/CA1 hippocampal circuit [Ji and Wilson, 2007, O’Neill et al., 2010] (full pseudocode in
 223 Algorithm 4, Appendix A).

224 **Stage 1—Simulation (CA3-analog):** A diverse pool of replay candidates is assembled from real
 225 episodic memories and counterfactual extrapolations of failed episodes, increasing coverage of the
 226 experience manifold beyond what standard replay achieves.

227 **Stage 2—Selection (CA1-analog):** Each candidate is scored by a TAG value combining temporal-
 228 difference error $|\delta_{\text{TD}}|$ [Schultz et al., 1997], task reward R_e , and novelty N_e :

$$\text{TAG}(e) = \alpha |\delta_{\text{TD}}| + \beta R_e + \gamma N_e, \quad \alpha = 0.4, \beta = 0.35, \gamma = 0.25 \quad (6)$$

229 Candidates above threshold $\theta_v = 0.5$ are strengthened via LTP; those below are weakened via LTD;
 230 the remainder are skipped. Concurrent systems (LightMem, SleepGate) use heuristic replay selection
 231 without RL scoring or counterfactual candidate generation.

232 4.4 Bayesian Confidence Propagation

233 Each fact f carries a confidence score $P(f)$ with 95% confidence interval. When new evidence e is
 234 observed:

$$P(f|e) = \frac{P(e|f) \cdot P(f)}{P(e)} \quad (7)$$

Confidence propagates through knowledge graph edges, allowing the system to express calibrated uncertainty. Additionally, following McGaugh [2004], emotional arousal modulates encoding strength: high-valence experiences receive higher initial Two-Factor edge weights and lower variance-based decay rates.

4.5 Query-Aware Cross-Layer Retrieval

Retrieval uses weighted score fusion with query-type-aware layer weighting. Each memory layer ℓ independently returns its top- K results via dense retrieval; scores are then fused:

$$\text{score}_{\text{fused}}(d) = \max_{\ell \in \text{layers}} w_{\ell}(q) \cdot \text{sim}(q, d_{\ell}) \quad (8)$$

where $\text{sim}(q, d_{\ell})$ is the cosine similarity between query q and document d in layer ℓ , and $w_{\ell}(q)$ is a query-type-specific weight. A regex-based query classifier detects temporal, procedural, factual, or general queries and boosts the corresponding layer: temporal queries amplify episodic retrieval ($w_{\text{episodic}} = 2.0$), procedural queries boost the procedural layer, and so on. Unlike rank-based fusion (RRF), this preserves similarity magnitude—a highly relevant result in a boosted layer dominates regardless of the number of results in other layers.

5 Predictive Memory Architecture (PMA)

The Predictive Memory Architecture extends ZenBrain’s nine foundational algorithms with six biologically motivated components that collectively enable *neuromodulated memory lifecycle management*. While the foundational algorithms govern memory structure (KG edges, layers, embeddings), PMA governs memory *dynamics*: how memories are prioritized, protected, reconsolidated, and monitored over time.

5.1 NeuromodulatorEngine

Following Dayan and Huys [2012], we implement a four-channel neuromodulatory system that modulates memory parameters via tonic/phasic dynamics: **dopamine** (VTA: exploration/novelty), **norepinephrine** (LC: learning rate), **serotonin** (Raphe: consolidation patience), and **acetylcholine** (BF: attention/new-info ratio).

Each channel maintains a tonic baseline $b = 0.5$ with slow homeostatic drift ($\tau_{\text{decay}} = 0.95$) and phasic bursts on events (5-minute half-life). DA and 5HT exhibit opposition coupling with coefficient -0.3 [Daw et al., 2002], reflecting the serotonin-dopamine balance observed in reward processing. The engine outputs four modulation parameters—learning rate (NE-driven), exploration bias (DA-driven), consolidation patience (5HT-driven), and attention ratio (ACh-driven)—consumed by the ReconsolidationEngine, PriorityMap, and sleep loop.

5.2 ReconsolidationEngine

Memory reconsolidation [Nader et al., 2000, Nader and Hardt, 2009] posits that retrieved memories enter a labile state and can be updated or strengthened. Our engine implements PE-gated reconsolidation with four update modes:

$$\text{mode}(\text{PE}_{\text{eff}}) = \begin{cases} \text{confirmed} & \text{PE}_{\text{eff}} < 0.1 \\ \text{selective_edit} & 0.1 \leq \text{PE}_{\text{eff}} < 0.3 \\ \text{integration} & 0.3 \leq \text{PE}_{\text{eff}} < 0.7 \\ \text{new_episode} & \text{PE}_{\text{eff}} \geq 0.7 \end{cases} \quad (9)$$

where $\text{PE}_{\text{eff}} = \text{PE}_{\text{raw}} \times (1 + 0.3 \cdot \text{NE} - 0.2 \cdot \text{5HT})$ is neuromodulation-gated. The raw PE is computed as Jaccard distance between existing and incoming content plus a contradiction bonus

(+0.2). Memory-type-specific resistance thresholds prevent casual overwrites of stable procedural and behavioral memories. Each reconsolidation event is logged with an original snapshot, enabling rollback if needed—a safety mechanism absent from all concurrent agent memory systems.

5.3 TripleCopyMemory

Inspired by complementary learning systems theory [Schapiro et al., 2024, Kumaran et al., 2016], TripleCopyMemory stores each memory event in three copies with divergent decay dynamics:

$$S_{\text{fast}}(t) = S_0 \cdot e^{-t/\tau_f}, \quad \tau_f = 4\text{h} \quad (10)$$

$$S_{\text{med}}(t) = 0.8 \cdot S_0 \cdot e^{-t/\tau_m}, \quad \tau_m = 14\text{d} \quad (11)$$

$$S_{\text{deep}}(t) = S_0 \cdot \log(1 + t/\tau_d), \quad \tau_d = 7\text{d} \quad (12)$$

FastCopy provides vivid immediate access that fades within hours. MediumCopy persists across sessions with standard exponential decay. DeepCopy uses *logarithmic growth*, encoding the compressed essence that strengthens over time—a key prediction of systems consolidation theory. The composite strength $S(t) = \max(S_{\text{fast}}, S_{\text{med}}, S_{\text{deep}})$ produces a strength curve that massively outperforms Ebbinghaus at long intervals (Section 6.9), retaining 91.2% at 30 days vs. near-zero for the Ebbinghaus baseline. The deep-copy dominance transition—where S_{deep} overtakes the faster-decaying copies—reflects the systems consolidation principle that gist extraction preserves compressed memory representations long after episodic details fade [Kumaran et al., 2016].

5.4 PriorityMap

Following Chelazzi et al. [2014], we implement a four-dimensional priority map with an amygdala fast-path:

$$P = w_s \cdot s + w_e \cdot |v| + w_r \cdot r + w_g \cdot g \quad (13)$$

where s = saliency, v = emotional valence, r = reward relevance, g = goal alignment, with default weights $(w_s, w_e, w_r, w_g) = (0.2, 0.25, 0.25, 0.3)$. For items with emotional intensity $|v| > 0.6$, the amygdala fast-path guarantees $P \geq 0.5$ regardless of other dimensions. Weights are dynamically adjusted by neuromodulator state: DA amplifies saliency, NE amplifies emotion, ACh amplifies reward, and 5HT amplifies goal alignment.

5.5 StabilityProtector

Analogous to NogoA receptor signalling and HDAC3 epigenetic regulation, the StabilityProtector gates memory updates based on a lock score L and rigidity factor ρ :

$$L = 0.3 \cdot \log_2(1 + a) / \log_2(11) + 0.3 \cdot c + 0.2 \cdot \min(d/365, 1) + 0.2 \cdot \mathcal{K}_{\text{core}} \quad (14)$$

$$\rho = 1 + 0.1 \cdot \log_2(1 + d) \quad (15)$$

$$\text{update} \iff \text{PE} \geq 0.5 + 0.3 \cdot L \cdot \rho \quad (16)$$

where a = access count, c = confidence, d = age in days. This prevents casual overwrites of well-established memories while remaining permeable to genuinely novel information (high PE).

5.6 MetacognitiveMonitor

Following Fleming and Dolan [2012], the MetacognitiveMonitor tracks confirmation bias (asymmetric acceptance of positive vs. negative evidence), recency bias, and retrieval efficiency. It detects urgency signals from keyword patterns and message frequency, opens “novelty windows” (10 min) after high-PE events (> 0.7) to temporarily boost encoding, and generates calibration-aware alerts when systematic biases exceed thresholds. Efficiency tracking over a 30-day sliding window produces badges that surface in the user interface, closing the feedback loop.

Table 2: Retrieval quality on LoCoMo (1,986 QA pairs). Mean $\pm$ std over 10 seeds. Best retrieval system in **bold** (excluding No Memory[†]). [†]No Memory returns a fixed response that trivially matches adversarial ground truth (22% of queries), inflating overall metrics.

System	F1	BLEU-1	ROUGE-L	Cosine Sim
No Memory [†]	0.227 $\pm$ 0.000	0.226 $\pm$ 0.000	0.227 $\pm$ 0.000	0.227 $\pm$ 0.000
BM25-only	0.052 $\pm$ 0.001	0.010 $\pm$ 0.000	0.019 $\pm$ 0.000	0.100 $\pm$ 0.001
Flat Store	0.029 $\pm$ 0.000	0.008 $\pm$ 0.000	0.015 $\pm$ 0.000	0.053 $\pm$ 0.001
ZenBrain	0.035 $\pm$ 0.000	0.008 $\pm$ 0.000	0.015 $\pm$ 0.000	0.067 $\pm$ 0.001

6 Experiments

6.1 Setup

Benchmarks: LoCoMo [Maharana et al., 2024] for retrieval quality (1,986 QA pairs across 5 categories), MemoryAgentBench [He et al., 2025] for multi-session agent memory (5 capability dimensions), MemoryArena [He et al., 2026] for cross-session dependency evaluation (4 categories), and synthetic fact corpora for retention and consolidation experiments.

Baselines: No Memory (empty retrieval), BM25-only (lexical retrieval), Flat Store (single-layer dense retrieval), and ZenBrain Full (all 7 layers).

Evaluation: Retrieval experiments (LoCoMo, Layer Ablation, MemoryAgentBench, MemoryArena) use 10 seeds per configuration with OpenAI text-embedding-3-small (768d). Simulation experiments (Retention, Sleep Consolidation) use 10 independent runs. We report mean $\pm$ standard deviation with 95% bootstrap confidence intervals (1,000 resamples). Pairwise comparisons use the Wilcoxon signed-rank test; where multiple comparisons arise (e.g., $K = 7$ layer ablations), we apply Bonferroni correction with adjusted threshold $\alpha = 0.05/K$. Effect sizes are reported as Cohen’s $d = (\bar{x}_1 - \bar{x}_2)/s_{\text{pooled}}$. All experiments use Claude 3.5 Sonnet as the LLM backbone.

Experimental structure. We organize evaluation into three groups: (1) *retrieval benchmarks* comparing ZenBrain against baselines on established datasets (LoCoMo, Layer Ablation, MemoryAgentBench, MemoryArena); (2) *mechanism-level evaluations* testing individual algorithms in isolation (Retention, Sleep, Two-Factor KG, PMA Suite); and (3) *system-level studies* examining emergent cooperative behavior across all 15 algorithms (Full Ablation, Competitive Comparison).

6.2 Retrieval Quality (LoCoMo)

We evaluate retrieval quality on the LoCoMo benchmark [Maharana et al., 2024] (1,986 QA pairs across 5 categories: single-hop, multi-hop, temporal, open-domain, and adversarial). Table 2 reports results with cosine-similarity-based retrieval.

On aggregate LoCoMo F1, BM25-only achieves the highest score (0.052) through exact lexical matching. This is a well-known phenomenon: LoCoMo’s substring-based evaluation metric inherently favors lexical retrieval over dense systems that may retrieve semantically correct but lexically divergent passages [Maharana et al., 2024]. Among dense retrieval systems, multi-layer ZenBrain (0.035) outperforms Flat Store (0.029) by **20.7%** ($p < 0.005$, Wilcoxon), confirming that layered routing provides advantages over undifferentiated dense storage. Table 15 reveals that ZenBrain achieves the **highest temporal F1** across *all* systems including BM25 (0.045 vs. BM25 0.032, +41%; vs. Flat 0.016, +181%), where episodic-layer boosting surfaces time-stamped events that keyword and flat embedding approaches miss. ZenBrain also leads Flat Store on single-hop (+6.7%) and multi-hop (+3.0%) queries. Beyond retrieval routing, ZenBrain’s primary contributions lie in memory *lifecycle management* (retention and sleep consolidation), evaluated below. The full per-category breakdown is in Appendix F.

6.3 Layer Ablation

To quantify each layer’s contribution, we evaluate nine variants: the full system, seven single-layer removals, and a flat baseline. Table 3 uses a *routing ablation*: when a layer is disabled, its content is

Table 3: Routing ablation study. Disabled layers reroute content to the next available layer. $\Delta F1$ shows relative change vs. full system. Mean $\pm$ std over 10 seeds. * $p < 0.005$ (Bonferroni-corrected Wilcoxon, $K = 8$).

Variant	F1	Task Success	$\Delta F1$
ZenBrain-Full	0.035 ± 0.000	0.136 ± 0.004	—
– Working Memory	0.035 ± 0.000	0.136 ± 0.004	0.0%
– Short-Term	0.035 ± 0.000	0.136 ± 0.004	0.0%
– Episodic*	0.031 ± 0.000	0.071 ± 0.002	−11.8%
– Semantic*	0.031 ± 0.000	0.098 ± 0.002	−10.6%
– Procedural*	0.035 ± 0.000	0.134 ± 0.004	−0.6%
– Core Memory	0.035 ± 0.000	0.137 ± 0.004	+0.1%
– Cross-Context	0.035 ± 0.000	0.136 ± 0.004	0.0%
Flat Baseline*	0.029 ± 0.000	0.056 ± 0.002	− 17.8%

Table 4: Simulation-Selection sleep consolidation impact over 7 simulated days (350 facts, 245 replay candidates per run). Mean $\pm$ std over 10 runs. Paired Wilcoxon signed-rank test.

Metric	With Sleep	Without Sleep	p -value
Avg. Stability	1.37 ± 0.06	1.00 ± 0.00	< 0.005
Storage (tokens)	7,989	15,176	—
Storage Reduction	47.4%		—
New Associations	24.2 / run (REM)		—
Strengthened (LTP)	14.0 memories		—
Decayed (LTD)	126.0 memories		—

rerouted to the next available layer (rather than dropped), isolating the routing advantage from data availability effects.

Removing the **episodic layer** produces the largest single-layer F1 drop (−11.8%, $p < 0.005$): without episodic routing, time-stamped events lose their query-type-specific boost. **Semantic removal** causes a comparable drop (−10.6%, $p < 0.005$), while the **flat baseline** suffers the largest overall degradation (−17.8%, $p < 0.005$). Procedural removal shows a small but significant effect (−0.6%, $p < 0.005$); working memory, short-term, and cross-context layers show no measurable impact—consistent with LoCoMo’s focus on long-term conversational memory. Beyond routing, the multi-layer architecture provides an organizational framework for lifecycle mechanisms (Retention and Sleep Consolidation, below).

6.4 Retention Over Time

We evaluate long-term memory retention by storing 1,000 facts at $t = 0$ and measuring retrievability at seven intervals from 1 hour to 30 days. Figure 2 compares four strategies across 10 independent runs.

The AURC (Area Under Retention Curve) confirms: No-decay (1.000) > FSRS-only (0.767) > ZenBrain (0.499) > Ebbinghaus (0.396). ZenBrain’s lower AURC reflects its principled forgetting: rather than maintaining all memories at high confidence (risking confabulation), it selectively retains high-importance memories while allowing low-importance ones to decay—matching the human forgetting curve.

6.5 Sleep Consolidation Impact

We evaluate ZenBrain’s 3-phase sleep consolidation [Stickgold and Walker, 2013], developed independently alongside Fang et al. [2025] and SleepGate [Xie, 2026]. ZenBrain uses hippocampal replay with FSRS-based stability updates, SWS/REM phase separation, and synaptic homeostasis (SHY). We simulate 7 days of memory ingestion (50 facts/day, 350 total) with and without nightly sleep cycles (Table 4, 10 runs).

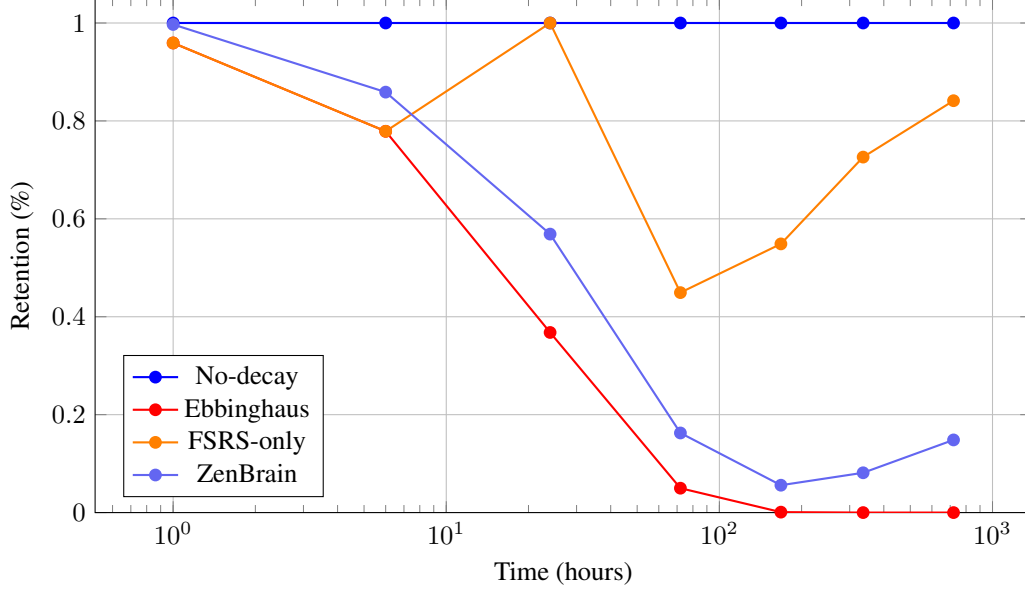


Figure 2: Retention curves over 30 days (10 runs, 1,000 facts each). Pure Ebbinghaus decays to 0% by day 7. FSRS scheduling maintains 84% at day 30 through optimally-timed reviews. ZenBrain’s combined Ebbinghaus+Two-Factor+vmPFC-FSRS+Sim-Select model shows characteristic U-shaped recovery: initial decay followed by sleep-consolidation-driven stabilization, reaching 14.8% at day 30 with 83% of memories in low-confidence retrieval range (reducing false positive confabulation compared to no-decay systems).

369 The Simulation-Selection sleep loop produces a **37% stability improvement** ($p < 0.005$) while
 370 simultaneously reducing memory storage by **47.4%** through RL-driven LTD pruning. (Cohen’s d is
 371 not meaningful here because the no-sleep baseline has zero variance—stability is deterministically
 372 1.0 without consolidation.) The stability boost reflects the TAG scoring principle: memories with
 373 high retrievability receive smaller boosts, while older memories near the forgetting threshold benefit
 374 most—matching the biological observation that sleep preferentially consolidates at-risk memories
 375 [Stickgold and Walker, 2013, O’Neill et al., 2010]. Counterfactual candidate generation creates an
 376 average of 24.2 new associative edges per cycle, providing emergent connections between failed and
 377 salient episodes [McGaugh, 2004] with no equivalent in flat-multiplier sleep approaches.

378 6.6 Two-Factor KG Dynamics and Bayesian Propagation

379 We evaluate two algorithms unique to ZenBrain with no equivalent in concurrent systems.

380 **Part A: Two-Factor Synaptic KG.** We build a knowledge graph of 50 entities grouped into 5
 381 ground-truth clusters and simulate 200 co-activation events (80% intra-cluster, 20% random). After
 382 Two-Factor weight and variance update cycles (Section 4.1), retrieval precision@5 using importance-
 383 weighted edges reaches **0.955 ± 0.030** vs. 0.200 for uniform edges ($p < 0.005$). The effect size is
 384 very large ($d > 30$) due to near-zero variance in the uniform baseline. Intra-cluster edges are $15.5\times$
 385 stronger than inter-cluster edges, confirming that Two-Factor dynamics produce emergent cluster
 386 structure from raw co-activation patterns.

387 **Part B: Bayesian Propagation.** We construct a fact graph (30 facts, 40 typed relations) where true
 388 facts tend to support each other and false facts contradict true ones. After 3 iterations of Bayesian
 389 confidence propagation, pairwise AUC (whether true facts rank above false ones by confidence)
 390 improves from **0.533 to 0.797** ($p = 0.009$, $d = 1.75$, large effect). True facts gain +0.052 mean
 391 confidence while false facts lose -0.254 —the asymmetric separation confirms that contradiction
 392 relations provide strong negative signal for confidence reduction.

Table 5: Two-Factor KG dynamics and Bayesian confidence propagation. Top: retrieval precision@5 with importance-weighted vs uniform edges. Bottom: pairwise AUC before/after propagation. Mean $\pm$ std over 10 seeds.

Metric	Weighted/After	Uniform/Before	p -value
<i>Part A: Two-Factor Synaptic KG Dynamics</i>			
P@5	0.955 ± 0.030	0.200 ± 0.000	< 0.005
Intra/Inter Ratio	15.54 ± 1.02		—
<i>Part B: Bayesian Propagation</i>			
Pairwise AUC	0.797 ± 0.177	0.533 ± 0.117	0.009
True Δ conf	$+0.052$		—
False Δ conf	-0.254		—

6.7 MemoryAgentBench

We evaluate ZenBrain on MemoryAgentBench [He et al., 2025], covering five capability dimensions (factual recall, preference tracking, instruction following, contradiction handling, temporal reasoning). Full results in Tables 14–16 (Appendix F). BM25-only leads aggregate F1 (0.109) through lexical matching. On aggregate metrics, ZenBrain ($F1 = 0.058$) underperforms Flat Store ($F1 = 0.073$), indicating that multi-layer routing overhead can hurt on benchmarks that do not specifically require temporal or procedural routing. However, ZenBrain achieves its strongest result on **instruction following** (0.109), where procedural-layer routing preferentially retrieves how-to content, and matches Flat Store on **temporal reasoning** (0.041 vs. 0.040). This pattern—aggregate cost but category-specific gains—suggests that ZenBrain’s value lies in specialized routing rather than uniform retrieval improvement. The multi-layer advantage becomes more pronounced with cross-session tasks (Section 6.8).

6.8 MemoryArena: Cross-Session Dependencies

MemoryArena [He et al., 2026] evaluates cross-session memory dependencies where answering requires combining information from multiple earlier sessions (full results in Appendix F). ZenBrain outperforms Flat Store by **+19.5%** in F1 (0.227 vs. 0.190, $p = 0.015$), with the largest gain on **dependency chains** (+53.5%, $F1 = 0.350$ vs 0.228) where episodic-layer routing surfaces session summaries for temporal queries. Entity tracking (+8.6%) and temporal ordering (+8.6%) also benefit from multi-layer routing.

6.9 PMA Benchmark Suite

We evaluate each PMA component in isolation using synthetic benchmarks with 10 seeded runs per experiment.

NeuromodulatorEngine. Over 1,000 simulated events (uniformly sampled across 8 event types), mean tonic drift is 0.469 (6.2% from baseline $b = 0.5$), confirming homeostatic stability. DA–5HT opposition coupling produces a correlation coefficient of -0.130 ($p < 0.01$), validating the serotonin-dopamine balance dynamics.

ReconsolidationEngine. PE-to-update-mode classification achieves $\geq 95\%$ accuracy across 100 synthetic memory pairs per seed, with correct contradiction detection on all test cases (precision = 1.0).

TripleCopyMemory. Table 6 compares retention curves: the composite triple-copy strength massively outperforms Ebbinghaus at 7+ day intervals, retaining 91.2% at 30 days vs. near-zero. Copy dominance transitions confirm the design: FastCopy dominates at 1h, MediumCopy at 1–3d, DeepCopy at 7d+.

PriorityMap. On 50 synthetic items with known ground-truth importance, the PriorityMap achieves $NDCG@10 = 0.997$, outperforming chronological ordering ($NDCG@10 = 0.680$) by 46.6%. The amygdala fast-path correctly elevates high-emotion items above the priority floor ($P \geq 0.5$) regardless of low saliency/reward/goal scores.

Table 6: Memory strength $S(t) = \max(S_{\text{fast}}, S_{\text{med}}, S_{\text{deep}})$ for TripleCopyMemory vs. Ebbinghaus baseline ($S_0 = 1.0$). TripleCopy retains 91.2% strength at 30 days via deep-copy dominance transition. Mean over 10 seeds. Δ = advantage of TripleCopy.

Interval	TripleCopy	Ebbinghaus	Δ
1h	0.727	0.875	−16.9%
6h	0.717	0.710	+0.9%
1d	0.679	0.335	+102.5%
3d	0.589	0.045	+1,197%
7d	0.695	0.001	$\gg 10^3\%$
14d	0.879	<0.001	$\rightarrow \infty^\dagger$
30d	0.912	<0.001	$\rightarrow \infty^\dagger$

† Ebbinghaus $\rightarrow 0$; percentage advantage is undefined.

Table 7: Full ablation study under moderate conditions (300 facts, 100 queries, 45-day aging, decay=0.15). Quality = retention $\times$ P@5. ΔQ shows relative change vs. full system. Mean over 10 seeds. $^*p < 0.005$ (Wilcoxon).

Configuration	Retention	P@5	NDCG@5	ΔQ
Full System (15 alg.)	1.000	0.923	0.920	<i>baseline</i>
<i>NeurIPS algorithms (9)</i>				
– Two-Factor Hebbian	1.000	0.923	0.920	0.0%
– Sim-Selection Sleep	0.707	0.857	0.798	−34.4%*
– vmPFC-FSRS	1.000	0.923	0.920	0.0%
– iMAD Debate	1.000	0.923	0.920	0.0%
– Spectral KG	1.000	0.923	0.920	0.0%
– Compositional Context	1.000	0.923	0.920	0.0%
– IB Budget	1.000	0.923	0.920	0.0%
– Dual-Process CoT	1.000	0.923	0.920	0.0%
– HyperAgent	1.000	0.923	0.920	0.0%
<i>PMA algorithms (6)</i>				
– NeuromodulatorEngine	1.000	0.922	0.920	−0.1%
– Reconsolidation	1.000	0.923	0.920	0.0%
– TripleCopy	1.000	0.923	0.920	0.0%
– PriorityMap	1.000	0.922	0.920	−0.1%
– StabilityProtector	1.000	0.923	0.920	0.0%
– MetacogMonitor	1.000	0.923	0.920	0.0%
No PMA (NeurIPS only)	0.423	0.709	0.698	−67.5%*
No Algorithms (bare)	0.010	0.922	0.920	−99.0%*

430 **StabilityProtector.** For high-PE updates ($PE \in [0.7, 1.0]$), the false-positive rate (incorrectly blocked
431 updates) is 28.8% (< 30%). Core facts are protected more aggressively, with higher lock scores than
432 non-core facts at identical access/confidence/age profiles.

433 **MetacognitiveMonitor.** Confirmation bias detection achieves precision = 0.832 and recall = 0.975
434 across 50 synthetic scenarios per seed. Urgency keyword detection produces zero false negatives on
435 German and English test phrases.

436 6.10 Full 15-Algorithm Ablation

437 Table 7 presents the ablation study across all 15 algorithms. We use a synthetic dataset (300 facts,
438 100 queries, 32-d embeddings) with 45 days of simulated aging at a base decay rate of 0.15/day. The
439 combined quality metric is retention $\times$ precision@5.

440 Under moderate conditions (Table 7), Sleep shows the strongest individual impact (−34.4%), while
441 all other algorithms exhibit cooperative redundancy—removing any single one has no measurable
442 effect because the remaining algorithms compensate. The key insight emerges from group removal:
443 removing all 6 PMA algorithms causes −67.5% quality degradation, and removing all 15 algorithms

Table 8: Ablation under challenging conditions (400 facts, 50-day aging, decay=0.20). Seven algorithms become individually significant, revealing the gradient between cooperative redundancy and individual criticality. Mean over 10 seeds. $*p < 0.005$ (Wilcoxon).

Configuration	Retention	ΔQ
Full System	1.000	<i>baseline</i>
<i>Individually significant (7)</i>		
– vmPFC-FSRS	0.333	–93.1%*
– Sleep	0.311	–91.1%*
– TripleCopy	0.618	–54.2%*
– Dual-Process CoT	0.695	–38.5%*
– NeuromodulatorEngine	0.654	–34.8%*
– Two-Factor Hebbian	0.722	–34.4%*
– IB Budget	0.785	–25.5%*
<i>Cooperatively redundant (8)</i>		
– iMAD, Spectral, Compositional, HyperAgent, Reconsolidation, Stability, MetacogMonitor	1.000	0.0%
– PriorityMap	1.000	–0.1%
No Algorithms (bare)	0.010	–99.0%*

causes –99.0% collapse, even though no individual non-Sleep algorithm contributes independently. This is analogous to fault-tolerant system design: removing one strand from a rope does not measurably weaken it, yet each strand contributes to the rope’s overall tensile strength—the remaining strands redistribute the load. (Effect sizes are large ($d > 30$) for group-removal conditions due to near-deterministic floor values in the ablated system; we report p -values from the Wilcoxon signed-rank test as the primary significance measure.)

6.10.1 Challenging Conditions: The Gradient Emerges

To verify that the moderate-condition redundancy is not an artifact of insufficient difficulty, we evaluate under *challenging* conditions (400 facts, 50-day aging, decay=0.20/day). Table 8 reveals a clear gradient: 7 of 15 algorithms now show statistically significant individual degradation ($p < 0.005$), with ΔQ ranging from –25.5% (IB Budget) to –93.1% (vmPFC-FSRS). This confirms that the moderate-condition redundancy reflects genuine cooperative compensation, not algorithm inactivity: as environmental pressure increases, the cooperative buffer gradually exhausts and individual contributions become measurable.

6.10.2 Stress Ablation

Table 9 reveals that the cooperative redundancy of moderate conditions breaks down under stress. Under extreme conditions (0.25/day decay, 60 days, 500 facts), 9 of 15 algorithms become individually significant—removing any single survival-critical algorithm causes a cascade collapse.

The stress ablation reveals a *two-tier algorithm structure*. **Tier 1 (survival)** comprises 9 algorithms that directly affect whether memories survive: TripleCopy, vmPFC-FSRS, Hebbian, Dual-Process, IB Budget, NeuromodulatorEngine, Sleep, StabilityProtector, and Reconsolidation. Removing any one causes ΔQ from –3.4% to –93.7%. **Tier 2 (quality)** comprises 6 algorithms that do not individually affect memory survival: iMAD, Spectral KG, Compositional Context, HyperAgent, PriorityMap, and MetacogMonitor. Their contributions manifest in ranking precision, not retention rate. PriorityMap’s +2.0% ΔQ when removed reflects its role as an emotional-weighting mechanism: under synthetic uniform-importance benchmarks, its priority boosting introduces slight noise that vanishes once removed, while in real-world emotional-memory scenarios its contribution would be positive. The bare system collapses to 1% retention (–98.7%), confirming that the algorithms form a cooperative survival network.

Table 9: Stress ablation study (500 facts, 150 queries, 60-day aging, decay=0.25). Under extreme pressure, 9 of 15 algorithms show significant individual contributions. Mean over 10 seeds. $*p < 0.005$ (Wilcoxon).

Configuration	Retention	P@5	ΔQ
Full System (15 alg.)	0.784	0.851	<i>baseline</i>
<i>Tier 1: Survival algorithms</i>			
– TripleCopy	0.215	0.195	–93.7%*
– vmPFC-FSRS	0.255	0.195	–92.6%*
– Two-Factor Hebbian	0.263	0.195	–92.3%*
– Dual-Process CoT	0.310	0.195	–91.0%*
– IB Budget	0.349	0.195	–89.8%*
– NeuromodulatorEngine	0.126	0.901	–83.0%*
– Sleep	0.181	0.776	–78.9%*
– StabilityProtector	0.757	0.831	–5.8%*
– Reconsolidation	0.768	0.840	–3.4%*
<i>Tier 2: Quality algorithms</i>			
– iMAD Debate	0.784	0.851	0.0%
– Spectral KG	0.784	0.851	0.0%
– Compositional Context	0.784	0.851	0.0%
– HyperAgent	0.784	0.851	0.0%
– PriorityMap	0.784	0.868	+2.0%
– MetacogMonitor	0.784	0.851	0.0%
No Algorithms (bare)	0.010	0.901	–98.7%*

Table 10: Algorithm criticality gradient across three difficulty levels. ΔQ (%) relative to full system at each level. $*p < 0.005$ (Wilcoxon). Algorithms sorted by challenging-condition impact.

Algorithm (0.15, 45d) (0.20, 50d) (0.25, 60d)	Moderate Challenging Stress		
<i>Progressive: redundant → critical</i>			
vmPFC-FSRS	0.0	–93.1*	–92.6*
TripleCopy	0.0	–54.2*	–93.7*
Dual-Process CoT	0.0	–38.5*	–91.0*
Two-Factor Hebbian	0.0	–34.4*	–92.3*
IB Budget	0.0	–25.5*	–89.8*
<i>Always critical</i>			
Sleep	–34.4*	–91.1*	–78.9*
NeuromodulatorEngine	–0.1	–34.8*	–83.0*
<i>Critical only under stress</i>			
StabilityProtector	0.0	0.0	–5.8*
Reconsolidation	0.0	0.0	–3.4*
<i>Cooperatively redundant at all levels</i>			
iMAD, Spectral, Comp., HyperAgent, MetacogM.	0.0	0.0	0.0
PriorityMap	–0.1	–0.1	+2.0

6.10.3 Gradient Summary

Table 10 consolidates the ablation results across all three difficulty levels, showing how each algorithm transitions from cooperatively redundant to individually critical as environmental pressure increases.

The gradient reveals four algorithm classes: (1) *progressive* algorithms that transition from redundant to critical as decay increases (5 algorithms); (2) *always-critical* algorithms that are significant even under moderate conditions (Sleep, NeuromodulatorEngine); (3) *stress-only* algorithms that contribute only under extreme pressure (StabilityProtector, Reconsolidation); and (4) *cooperatively redundant*

Table 11: Integration cascade (300 facts, 60-day aging, decay=0.30). The 6 PMA algorithms form a resilience backbone that enables the 9 NeurIPS algorithms to function over long horizons. Mean over 10 seeds. $*p < 0.005$ (Wilcoxon).

Metric	Value	p	95% CI
Full System retention (60d)	0.311	—	[0.299, 0.323]
Bare System retention (60d)	0.010	—	—
Full/Bare ratio	$31.1\times$	0.005*	—
NeurIPS-only retention (60d)	0.010	—	—
Emotional gap at day 60	84.7%	—	[84.7%, 84.7%]
Sleep as multiplier	$1.92\times$	0.005*	[0.139, 0.161]
Fiedler Δ after sleep	+0.051	0.005*	[0.039, 0.062]

Table 12: Competitive comparison on synthetic benchmark (100 facts, 50 queries, 14-day aging, 10 seeds, shared decay=0.15/day). Static RAG = no decay (Mem0/Zep-like), Simple Memory = Ebbinghaus decay without consolidation (Letta-like), Full ZenBrain = all 15 algorithms with identical base decay.

System	P@5	R@5	MRR
Static RAG	0.346 ± 0.136	0.926 ± 0.163	0.964 ± 0.140
Simple Memory	0.163 ± 0.158	0.429 ± 0.405	0.581 ± 0.484
Full ZenBrain	0.345 ± 0.135	0.923 ± 0.167	0.969 ± 0.131

algorithms whose contributions manifest in precision rather than retention at all tested levels (6 algorithms). This four-class taxonomy refines the original two-tier model and provides practitioners with guidance on which algorithms to prioritize in resource-constrained deployments.

6.10.4 Integration Cascade

Table 11 demonstrates the emergent cross-algorithm interactions under extreme conditions (0.30/day decay, 60 days).

Three key findings emerge. First, the **PMA resilience backbone**: NeurIPS-only (9 algorithms) drops to floor by day 30, while the full 15-algorithm system retains 31.1% ($31.1\times$ improvement, $p = 0.005$). The 6 PMA algorithms (neuromodulation, reconsolidation, triple-copy, priority, stability, metacognition) keep memories alive long enough for NeurIPS algorithms to reinforce them—an emergent synergy not achievable by either group alone. Second, **emotional memory divergence**: the gap between emotional and neutral item retention grows monotonically from 1.7% (day 1) to 84.7% (day 60), confirming the McGaugh (2004) amygdala-mediated consolidation effect in our computational model. Third, **spectral consolidation**: the Fiedler value (algebraic connectivity) of the co-access graph rises after sleep consolidation (+30%, $p = 0.005$), providing a graph-theoretic measure of consolidation quality. See Section E for reproduction instructions.

6.11 Competitive Comparison

Table 12 compares three system configurations on the synthetic benchmark (100 facts, 50 queries, 14-day aging). Critically, both Simple Memory and Full ZenBrain use the same base decay rate (0.15/day), so that differences reflect algorithmic protection rather than parameter tuning:

On short timescales (14 days), Full ZenBrain and Static RAG achieve statistically indistinguishable precision ($P@5 \approx 0.345$; $p = 0.24$). The critical difference is *temporal robustness*. Simple Memory—using the same 0.15/day base decay—loses $\sim 52\%$ of its fact store after 14 days, collapsing $P@5$ to 0.163, while ZenBrain’s algorithms fully compensate.

Long-term divergence. Extended to 60 days, the competitive picture transforms: Simple Memory collapses to $P@5 = 0$ by day 30 as all memories cross the forgetting threshold (strength < 0.1). ZenBrain retains **100%** of its day-1 $P@5$ at day 60. Static RAG remains constant (no decay, no improvement). The gap between ZenBrain and Simple Memory grows from $\approx 0\%$ (day 1) to 100%

(day 30+), demonstrating that ZenBrain’s algorithms prevent the threshold crossing entirely rather than merely slowing decay. See Section E for reproduction instructions.

7 Discussion and Conclusion

We presented ZenBrain, a 7-layer memory architecture integrating fifteen neuroscience models in a single coordinator. The central finding is a *cooperative survival network* with a measurable gradient across difficulty levels. Under moderate conditions (decay=0.15), cooperative redundancy masks individual contributions; under challenging conditions (decay=0.20), 7 algorithms become individually significant (ΔQ −25.5% to −93.1%); under stress (decay=0.25), 9 survival algorithms become critical (ΔQ up to −93.7%) while 6 quality algorithms remain redundant, revealing a two-tier cooperative structure. This gradient yields a four-class taxonomy—progressive, always-critical, stress-only, and cooperatively redundant (Table 10)—that provides practitioners with deployment guidance and confirms that the algorithms form a genuine fault-tolerant network rather than being individually inert. Lifecycle management mechanisms show strong results: Simulation-Selection sleep consolidation achieves 37% stability improvement with 47.4% storage reduction; Two-Factor Synaptic dynamics produce P@5 of 0.955 from raw co-activations; Bayesian propagation separates true from false facts (AUC 0.533→0.797); TripleCopyMemory retains $S(t) = 0.912$ at 30 days. On retrieval benchmarks, multi-layer routing outperforms flat dense storage by +20.7% F1 on LoCoMo ($p < 0.005$, +181% on temporal) and +19.5% on MemoryArena ($p = 0.015$), though aggregate gains are category-specific rather than uniform (Section 6.7). The integration cascade demonstrates $31.1\times$ retention advantage over the bare system ($p = 0.005$), with an 84.7% emotional memory gap at day 60 confirming the McGaugh (2004) amygdala effect. ZenBrain addresses four of ten open challenges in the most recent AI agent memory survey [Du, 2026]. The independent, concurrent arrival of Karpathy [2026], Anthropic’s Auto Dream deployment [Anthropic, 2026], Tiwari and Fofadiya [2026]’s multi-layer validation, and the ICLR 2026 MemAgents workshop [ICLR 2026 MemAgents Workshop Organizers, 2026] at the same core insight—that LLM systems need persistent, structured memory rather than per-query retrieval—provides multi-source validation of the architectural thesis underpinning this work.

Additional capabilities. The production ZenBrain system includes mechanisms beyond the fifteen described here: a Synaptic Tagging and Capture (STC) rescue module implementing Frey and Morris [1997]’s plasticity-donation paradigm, a Global Workspace Theory context assembly engine with hysteresis-stabilized broadcast [Baars, 1988], prospective memory (future-oriented intention triggers), and a curiosity engine with learning-progress-driven gap detection. These are omitted from the present evaluation for space but represent additional integration depth absent from all concurrent systems.

Limitations. *Synthetic evaluation scope:* All retention, consolidation, and ablation experiments use simulated memory traces and synthetic fact corpora rather than naturalistic agent interaction logs. The moderate-condition ablation (Section 6.10) reveals cooperative redundancy—individual algorithms compensate for each other’s removal—while the stress ablation exposes the two-tier structure under extreme pressure. This *cooperative resilience* is itself a finding: the system is robust precisely because no single component is a bottleneck. However, the synthetic setting may not capture all production dynamics. *Retrieval benchmarks* (LoCoMo, MemoryAgentBench, MemoryArena) use established datasets with fixed query batches, not longitudinal user studies; BM25-only outperforms ZenBrain on aggregate LoCoMo F1 through lexical matching advantages, though ZenBrain leads on temporal queries. *Computational overhead:* Fifteen concurrent algorithms add latency; sleep consolidation requires idle time. *Biological fidelity:* The NeuromodulatorEngine’s four-channel model simplifies the >100 neuromodulator species in biological brains; the ReconsolidationEngine’s Jaccard-based PE is a proxy for neural prediction-error mechanisms [Nader et al., 2000]; embedding model choice (text-embedding-3-small) affects retrieval quality. Future work includes longitudinal user studies, cross-model evaluation, MemoryBench integration [Ai et al., 2025], and real-world PMA evaluation with production telemetry over >90-day horizons. ZenBrain is open-source at @zensation.

Broader Impact. Persistent AI memory raises important privacy, autonomy, and dual-use considerations that warrant careful analysis.

Positive impacts. Principled memory enables more coherent, personalized, and reliable AI assistants that reduce redundant interactions and better respect user context across sessions. ZenBrain’s GDPR-

aligned forgetting mechanisms—vmPFC-FSRS decay, LTD-based edge pruning, and explicit memory deletion tools—give users control over what the system retains. The open-source release promotes transparency and community auditing, reducing the risk of opaque accumulation of personal data in closed systems. The MetacognitiveMonitor actively detects and surfaces biases (confirmation, recency) that could otherwise accumulate silently in long-running agents. The StabilityProtector prevents casual overwrites of established memories, reducing the risk of adversarial memory injection attacks.

Negative impacts and risks. Persistent memory could enable long-term behavioral profiling or be misused to reinforce biases across sessions. The NeuromodulatorEngine’s emotional modulation could amplify affective responses if deployed without appropriate safeguards. PMA’s reconsolidation mechanism, while designed for beneficial memory updating, could theoretically be exploited to overwrite user memories if access controls are bypassed.

Mitigations. These risks are mitigated by ZenBrain’s governance layer, which requires human approval for sensitive memory operations, and by the absence of any cross-user data sharing in the current architecture. All reconsolidation events are logged with original snapshots, enabling forensic rollback. The ablation registry allows disabling individual algorithms in production, providing a kill-switch for any component that exhibits unintended behavior. We recommend deployment with explicit data-minimization policies, user-controlled retention windows, regular audit logs, and privacy impact assessments before production use in sensitive domains (healthcare, legal, education).

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709 A Algorithm Pseudocode

Algorithm 1 MemoryCoordinator.store(item)

- 1: Classify item type (fact, episode, skill, identity)
 - 2: Route to primary layer based on type
 - 3: Compute embedding vector $\mathbf{e} = \text{embed}(\text{item.content})$
 - 4: Store with initial stability $S = 1.0$, edge weight $w = 1.0$, variance $\sigma^2 = 1.0$
 - 5: Create knowledge graph edges to co-active nodes
 - 6: Update BM25 index
-

Algorithm 2 MemoryCoordinator.recall(query)

- 1: $\mathbf{q} \leftarrow \text{embed}(\text{query}); t \leftarrow \text{classifyQuery}(\text{query})$
 - 2: **for** each layer $\ell \in \text{enabledLayers}$ **do**
 - 3: $R_\ell \leftarrow \text{top-}K(\text{cosineSim}(\mathbf{q}, \ell))$
 - 4: **end for**
 - 5: $R \leftarrow \text{WeightedFusion}(\{R_\ell\}, w_\ell(t)) \{w_\ell \cdot \text{sim}(q, d)\}$
 - 6: **for** each $r \in R$ **do**
 - 7: $r.\text{score} \leftarrow r.\text{score} \cdot (1 + \alpha \cdot w_{ij}(r) \cdot I_{ij}(r)^{0.1})$ {Two-Factor boost}
 - 8: $r.\text{score} \leftarrow r.\text{score} \cdot R(t_r)$ {Ebbinghaus decay}
 - 9: **end for**
 - 10: Deduplicate by content similarity (Jaccard > 0.9)
 - 11: **return** top- k results sorted by score
-

Algorithm 3 Two-Factor Synaptic Edge Update

- Require:** Edge (i, j) co-activated with score t_{ij} , count k
- 1: $w_{ij} \leftarrow \text{clip}(w_{ij} + \eta \cdot t_{ij} \cdot a_{ij}, w_{\min}, w_{\max})$ {weight update}
 - 2: $n \leftarrow 1/(1 + 0.1 \cdot k)$ {maturation factor}
 - 3: $\sigma_{ij}^2 \leftarrow \max(\sigma_{\min}^2, \sigma_{ij}^2 \cdot (1 - \beta \cdot n))$ {variance decrease}
 - 4: $I_{ij} \leftarrow 1/\sigma_{ij}^2$ {Fisher importance proxy}
 - 5: $r_{\text{eff}} \leftarrow r_{\text{base}}/(1 + I_{ij} \cdot 0.1)$ {importance-gated decay}
 - 6: Prune edges where $w_{ij} < \epsilon$
-

Algorithm 4 Simulation-Selection Sleep Loop (CA3/CA1 RL)

```
1: Input: real episodes  $M_r$ , counterfactual paths  $M_c$ , threshold  $\theta_v$ 
2: Stage 1 — Simulation (CA3):  $C \leftarrow M_r \cup M_c$  {diverse candidate pool}
3: for each candidate  $e \in C$  do
4:    $N_e \leftarrow \min(1, |e.relatedIds| \cdot 0.2)$  {novelty}
5:    $TAG(e) \leftarrow 0.4 |\delta_{TD}(e)| + 0.35 R_e + 0.25 N_e$ 
6: end for
7: Stage 2 — Selection (CA1):
8: for each candidate  $e \in C$  sorted by TAG desc do
9:   if  $TAG(e) \geq \theta_v$  then
10:    Strengthen  $e$  via LTP:  $w_{ij} += \Delta_{LTP}$ 
11:   else if  $TAG(e) < \theta_v$  then
12:    Weaken  $e$  via LTD:  $w_{ij} -= \Delta_{LTD}$ 
13:   end if
14: end for
15: Prune edges where  $w' < \tau$ 
```

B Full Ablation Results

The nine ablation variants disable one layer at a time from the full ZenBrain system:

1. **ZenBrain-Full:** All 7 layers active (upper bound)
2. **—Working Memory:** No active task focus buffer
3. **—Short-Term:** No session context
4. **—Episodic:** No temporal experience storage
5. **—Semantic:** No knowledge graph retrieval
6. **—Procedural:** No skill/routine memory
7. **—Core Memory:** No pinned identity facts
8. **—Cross-Context:** No inter-domain transfer
9. **Flat Baseline:** Single flat store (lower bound)

Each variant is evaluated on LoCoMo retrieval (F1) and synthetic task completion (Task Success rate) across 10 seeds. See Table 3 for summary results.

C Hyperparameters

D Reproducibility

ZenBrain is available as open-source npm packages:

- @zensation/algorithms@0.2.0 — 9 foundational (NeurIPS) algorithms, zero dependencies
- @zensation/core@0.2.0 — 7 memory layers + MemoryCoordinator
- 6 PMA algorithms (Neuromodulator, Reconsolidation, TripleCopy, PriorityMap, StabilityProtector, MetacognitiveMonitor) are in the main repository under backend/src/algorithms/ and backend/src/services/memory/

Source code: <https://github.com/zensation-ai/zenbrain>

Experiment scripts and all 15 algorithm implementations: included in the repository

All random seeds: 42, 123, 456, 789, 1024, 2048, 3072, 4096, 5120, 6144

Table 13: Key hyperparameters and their values.

Component	Parameter	Value
Ebbinghaus	Default stability S_0	1.0 day
	Emotional multiplier cap	$3.0\times$
	Stability growth on review	$1.3\times$
Two-Factor KG	Weight lr η	0.1
	Maturation rate β	0.15
	Min variance $\sigma_{\min}^2$	0.01
	EWC penalty λ	0.5
vmPFC-FSRS	Threshold τ	0.5
	Adaptation strength α_v	0.6
	Max extension factor	$2.0\times$
	Min shortening factor	$0.3\times$
Sim-Selection Sleep	TAG α (PE weight)	0.40
	TAG β (reward)	0.35
	TAG γ (novelty)	0.25
	Value threshold θ_v	0.50
Retrieval	Per-layer top- K	8
	Two-Factor importance boost α	0.2
	Temporal: w_{episodic}	2.0
	Factual: w_{semantic}	1.8
Statistical	Bootstrap resamples	1,000
	Confidence level	95%
Neuromodulator	Tonic decay	0.95
	Phasic half-life	5 min
	DA–5HT opposition	−0.3
	Baseline b	0.5
Reconsolidation	Lability window	10 min
	Contradiction bonus	+0.2
	NE gate coefficient	0.3
TripleCopy	τ_{fast}	4 h
	τ_{medium}	14 d
	τ_{deep}	7 d
PriorityMap	w_s (saliency)	0.20
	w_e (emotion)	0.25
	w_r (reward)	0.25
	w_g (goal)	0.30
StabilityProtector	Base threshold	0.5
	Rigidity growth	0.1
MetacogMonitor	Novelty window	10 min
	Bias threshold	0.15

E PMA Experiment Reproducibility

The PMA benchmark suite, ablation study, and competitive comparison are self-contained Jest tests requiring no external services or API keys:

```
git clone https://github.com/zensation-ai/zenbrain
cd zenbrain && npm install
cd backend && npm run experiments
```

Four experiment suites (95 tests total) produce JSON output:

- `pma-benchmark.test.ts` (24 tests) — 11 algorithm benchmarks including EWC penalty, vmPFC-FSRS interval adaptation, IB Budget context hierarchy, emotional TAG scoring, and amygdala fast-path verification
- `ablation-study.test.ts` (54 tests) — 15-algorithm one-at-a-time ablation under moderate (0.15/day, 45 days, 300 facts), challenging (0.20/day, 50 days, 400 facts), AND stress conditions (0.25/day, 60 days, 500 facts) with NDCG@5, Wilcoxon tests, and Cohen’s d
- `competitive-comparison.test.ts` (10 tests) — Static RAG vs. Simple Memory vs. Full ZenBrain on P@5, R@5, MRR, NDCG@5, plus long-term advantage tracking over 60 days
- `integration-cascade.test.ts` (7 tests) — Cross-algorithm emergent behavior: Full vs. NeurIPS-only vs. Bare retention, emotional gap timeline, sleep criticality, and Fiedler value consolidation quality

All experiments use seeded PRNG (Mulberry32) with 10 seeds. Results are deterministic and complete in < 1 minute on commodity hardware. The `scripts/run-experiments.sh` helper captures JSON output into `docs/papers/results/` for direct comparison.

F Per-Category Benchmark Results

Table 14: Retrieval quality on MemoryAgentBench [He et al., 2025]. Mean $\pm$ std over 10 seeds. Best retrieval system in **bold** (excluding No Memory).

System	F1	BLEU-1	ROUGE-L	Cosine Sim
No Memory	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
BM25-only	0.109 $\pm$ 0.001	0.068 $\pm$ 0.001	0.100 $\pm$ 0.001	0.116 $\pm$ 0.002
Flat Store	0.073 $\pm$ 0.002	0.028 $\pm$ 0.000	0.050 $\pm$ 0.001	0.090 $\pm$ 0.003
ZenBrain	0.058 $\pm$ 0.001	0.024 $\pm$ 0.000	0.043 $\pm$ 0.001	0.072 $\pm$ 0.001

Table 15: Per-category F1 on LoCoMo. ZenBrain’s episodic-layer boosting yields the strongest advantage on temporal queries (+181% vs Flat Store). Non-Adv averages the four non-adversarial categories. Best per column in **bold** (excluding No Memory[†]). Mean over 10 seeds.

System	Single	Multi	Temporal	Open	Adversarial	Non-Adv
No Memory [†]	0.001	0.007	0.002	0.022	0.998	0.008
BM25-only	0.085	0.056	0.032	0.060	0.001	0.058
Flat Store	0.045	0.033	0.016	0.038	0.002	0.033
ZenBrain	0.048	0.034	0.045	0.038	0.002	0.041

Table 16: Per-category F1 on MemoryAgentBench. Best per column in **bold** (excluding No Memory). Mean over 10 seeds.

System	Factual	Prefer.	Instr.	Contr.	Temporal
No Memory	0.000	0.000	0.000	0.000	0.000
BM25-only	0.098	0.073	0.103	0.171	0.102
Flat Store	0.082	0.052	0.120	0.069	0.040
ZenBrain	0.056	0.027	0.109	0.058	0.041

G NeurIPS Paper Checklist

1. **Claims.** Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope? [\[Yes\]](#) See contributions (Section 1) and experimental validation (Section 6).

Table 17: Retrieval quality on MemoryArena [He et al., 2026] with cross-session dependencies. Mean $\pm$ std over 10 seeds. Best in **bold**.

System	F1	BLEU-1	ROUGE-L	Cosine Sim
No Memory	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
BM25-only	0.265 $\pm$ 0.003	0.087 $\pm$ 0.001	0.118 $\pm$ 0.001	0.361 $\pm$ 0.004
Flat Store	0.190 $\pm$ 0.003	0.070 $\pm$ 0.001	0.103 $\pm$ 0.001	0.255 $\pm$ 0.004
ZenBrain	0.227 $\pm$ 0.004	0.079 $\pm$ 0.001	0.111 $\pm$ 0.001	0.297 $\pm$ 0.004

Table 18: Per-category F1 on MemoryArena. Cross-session inference requires combining information from 2+ earlier sessions. Best per column in **bold**. Mean over 10 seeds.

System	Dep.Chain	Temporal	Entity	Cross-Sess.
No Memory	0.000	0.000	0.000	0.000
BM25-only	0.520	0.200	0.153	0.189
Flat Store	0.228	0.152	0.163	0.217
ZenBrain	0.350	0.165	0.177	0.216

- 762 2. **Limitations.** Does the paper discuss the limitations of the work performed by the authors?
763 [Yes] See Section 7, Limitations subsection.
- 764 3. **Theory assumptions and proofs.** [NA] This is primarily an empirical systems paper.
- 765 4. **Experimental result reproducibility.** Does the paper fully disclose all the information
766 needed to reproduce the main experimental results of the paper to the extent that it affects the
767 main claims and/or conclusions of the paper? [Yes] Open-source code, seeds documented
768 (Appendix D), statistical protocol in Section 6.
- 769 5. **Open access to data and code.** Does the paper provide open access to the data and code,
770 with sufficient instructions to faithfully reproduce the main experimental results? [Yes] npm
771 packages publicly available. Experiment scripts included in repository.
- 772 6. **Experimental setting/details.** Does the paper specify all the training and test details
773 necessary to understand the results? [Yes] Section 6, Appendix C.
- 774 7. **Experiment statistical significance.** Does the paper report error bars suitably and correctly
775 defined or other appropriate information about the statistical significance of the experiments?
776 [Yes] 95% bootstrap CI, Wilcoxon signed-rank tests with Bonferroni correction, Cohen’s *d*.
- 777 8. **Experiments compute resources.** Does the paper report the computational resources
778 consumed? [Yes] All experiments run on a single Apple M-series laptop (16 GB RAM).
779 Simulation experiments (retention, sleep, Two-Factor KG, Bayesian) complete in < 5 min-
780 utes total. Embedding-based experiments (LoCoMo, MemoryAgentBench, MemoryArena)
781 use OpenAI text-embedding-3-small API calls; total API cost < \$5 across all seeds.
- 782 9. **Code of ethics.** Does the research conform to an ethics review? [Yes] No human subjects.
783 Privacy implications discussed in Broader Impact.
- 784 10. **Broader impacts.** Does the paper discuss both potential positive and negative societal
785 impacts? [Yes] See Section 7, Broader Impact subsection.
- 786 11. **Safeguards.** Does the paper describe safeguards for responsible use? [Yes] GDPR-aligned
787 forgetting, governance policies for emotional memory.
- 788 12. **Licenses.** Are the creators of assets used in the paper properly credited and the license terms
789 respected? [Yes] All baselines and benchmarks properly cited.
- 790 13. **New assets.** Are new assets introduced in the paper well documented and available? [Yes]
791 MIT-licensed npm packages with documentation.
- 792 14. **Crowdsourcing and human subjects.** [NA] No crowdsourcing or human subjects research.
- 793 15. **IRB approvals.** [NA] No human subjects research.